

YOONMIN LEE

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EDUCATION

University of British Columbia

Bachelor of Applied Science - Mechanical Engineering (Mechatronics Option)

Vancouver, BC

Expected Graduation: May 2028

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, AutoCAD, Onshape, STAR-CCM+

Manufacturing & Fabrication: FDM 3D Printing, Machining (Mill, Lathe), Hand Tools

Programming: MATLAB, Python, C/C++

Instrumentation & Electronics: Soldering, Oscilloscope, Multimeter, Arduino

EXPERIENCE

Aerodynamics Designer | UBC Supermileage (Vancouver, BC)

Sep. 2025 – Present

- Created aerodynamic vehicle shell designs using surface modelling techniques in SolidWorks, ensuring designs meet manufacturing and assembly constraints
- Explored vehicle concepts focused on ergonomic design, providing insight into the trade-offs between aerodynamic performance and driver comfort
- Developed the team's 3D modelling workflow and standardized the team's surface modelling techniques used to connect multiple surfaces flawlessly, improving simulation results and aerodynamic performance
- Conducted Computational Fluid Dynamics (CFD) validation studies in STAR-CCM+, and established the team's new CFD methodology by performing a baseline validation simulation, ensuring that the template results correlate with reality
- Created the base SolidWorks assembly template for the team's Urban vehicle concept, allowing members to conceptualize shell designs while adhering to vehicle constraints and maintaining consistency across designs
- Engineered an adjustable, parametric human body model in SolidWorks by modifying joint geometry and global equations, enabling driver dimension inputs for custom ergonomic packaging and driver safety compliance

3D Modeller | UBC BIOMOD (Vancouver, BC)

Oct. 2024 – Present

- Developed and created animations of 3D molecular models using ChimeraX and Blender
- Applied science communication and storytelling principles to clearly explain complex scientific concepts to non-technical audiences through animated videos
- Placed third place for video presentation at the 2024 BIOMOD Annual Competition

PROJECTS

Water Delivery Vehicle | MECH 223 Team Project

Jan. 2026 – Mar. 2026

- Worked in a group of seven to design a remote-controlled vehicle capable of extinguishing small fires
- Designed and fabricated a mechanism consisting of sheet metal and servos, allowing the driver to control the movement the vehicle's water aiming system
- Modeled and 3D printed custom wheel-axle attachment components to improve torque transfer
- Conducted a focused experiment to determine frictional losses in a water tube and nozzle system, and analysed experimental data in MATLAB to optimize the system's specifications
- Simulated the nozzle discharge trajectory in MATLAB to determine the amount of time spent on extinguishing fires, improving vehicle performance and driver confidence
- Improved overall vehicle performance and reliability by applying FMEA to identify and mitigate risks
- Constructed a comprehensive Bill of Materials (BOM) to track fabrication costs, ensuring full compliance with project budget constraints.

BAL-I | UBC EDT Competition

Mar. 2026

- Conceptualized and designed 3D CAD models for an autonomous trash-collecting robot
- Investigated existing environmental gaps and constraints to justify technical requirements and design decisions
- Collaborated in a 3-person team using Onshape to model a multi-component mechanical assembly

Study-Gotchi | Python

Dec. 2025 – Mar. 2026

- Built a Python game using Pygame, where users study to take care of a pet octopus; The user's study time is recorded and converted into in-game currency
- Implemented a Pomodoro-style timer to promote efficient study habits

The Mag-Lev | Soldering, Machining Tools

Aug. 2025 – Sep. 2025

- Fabricated a magnetic levitation device by soldering electronic components and operating machining tools
- Practiced careful machining techniques on the lathe, mill, and drill press, achieving results within a given tolerance
- Soldered a PCB board containing capacitors, LEDs, and resistors